GWCD Documentation - Geocoding

Vermont AHS GIS and Data Viz | Groundwater Chemistry Database (GWCD)

Geocoding: Geocoded vs. Non-Geocodable Data

Process

1. Each record's address is concatenated (`Address1, City, VT, Zip`) and geocoded against the state's E911 composite address locator.
2. Any record whose address **fails** to geocode is retried using **just the town name**, geocoded to that town's **centroid** point instead.
3. Only a record that fails **both** steps (no usable address, and no recognizable/geocodable town name) is dropped from the geocoded/mapped product.
4. All surviving points (address-level matches + centroid fallbacks) are spatially joined to town boundaries to assign a `TOWNNAME`.

Practical implication: because of the centroid fallback, the overwhelming majority of records end up with a town assignment even when the home address itself can't be pinpointed. Geocoding failure (a record with no `TOWNNAME` at all) should be rare — it requires the town/city field itself to also be unusable (blank, out-of-state, unresolvable typo not covered by the alias table, etc.).

Is there a single view with both geocoded and non-geocodable results?

Not as a single ready-made table, but it's straightforward to assemble:

Table	Geocoded?	Filtered to private-water scope?	Use for
`GroundwaterChemistry_GWCDrawdata`	No	No	**Full, unfiltered dataset** — best for state-level totals/rates that don't need geometry or town-level slicing
`GroundwaterChemistry_GWCDPoints`	Yes (address or town centroid)	Yes	Town-level slicing with unsuppressed counts — best source for a grant-reporting dashboard
`GroundwaterChemistry_GWCDPointTableExternal`	Yes	Yes, plus deduplicated + suppressed (<20 samples/group hidden)	What the public app displays — don't use this for totals, you'll undercount

If you need one dataset with **both** full records and a town flag, join `GroundwaterChemistry_GWCDrawdata` to `GroundwaterChemistry_GWCDPoints` on `Record_uid` — this gives every raw record, with a `TOWNNAME` where one could be determined and NULL where it couldn't.

For a dashboard that needs **both** state-level and town-level views: start from `GroundwaterChemistry_GWCDrawdata` (or the raw SQL source) for the state-level numbers — it needs no geocoding — and use `GroundwaterChemistry_GWCDPoints`'s `TOWNNAME` field for the town-level slice.